PERSONAL INFORMATION

Md. Sadaab Hassan



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STUDIES APPLIED FOR

Erasmus Mundus Master Degree in Plant Health in Sustainable Cropping Systems (Plant Health)

RESEARCH EXPERIENCE

01/10/2024 – Present

Research Associate

Molecular Biology & Protein Science Laboratory, University of Rajshahi.

Major Tasks: Recombinant protein expression, Molecular analysis for sustainability, Pest diagnosis using markers, Bio-based pest solutions, Lab management and training etc.

01/06/2023–30/09/2024

Project Student

Molecular Health Science Laboratory, University of Rajshahi.

Major Tasks: Pathogen identification and characterization, Food safety evaluation, Sustainable disease management, Impact assessment on exports, Interdisciplinary collaboration, and Research dissemination etc.

05/01/2023–Present

Assistant Lab Manager

Advanced Bioinformatics, Computational Biology and Data Science Laboratory, Bangladesh (ABCD Laboratory, Bangladesh).

Major Tasks: Data management and analysis, Bioinformatics application, Lab operations coordination, and Scientific reporting etc.

01/07/2022–Present

Research Assistant

ABCD Laboratory, Bangladesh.

Major Tasks: Natural plant-based drug design, Plant health modelling, Data visualization, and Sustainable Agriculture research etc.

01/11/2021– 31/05/2023

Research Intern

Molecular Pathology Laboratory, Institute of Biological Science, University of Rajshahi.

Major Tasks: Laboratory experimentation, Data collection and analysis, Eco-friendly solutions, and Research communication etc.

01/01/2021-01/06/2022

Junior Research Collaborator

ABCD Laboratory, Bangladesh.

Major tasks: Natural product research, Bioinformatics tool utilization, Data integration for research, Collaborative research support, and Research documentation etc.

RESEARCH PROJECTS

05/11/ 2025 – Current

Predicting Antimicrobial Resistance Genes in Virulent *Salmonella Typhi* Strains Using Integrated Genome Sequencing and Systems Biology

12/09/ 2025 – Current

Identification of Key Pest Gene Network Modules and Hub Genes Coupled with Molecular Dynamics Simulations for Selective IPM Agrochemicals

EDUCATIONAL BACKGROUND

28/10/2024 – Present

Master of Science in Genetic Engineering and Biotechnology (MS in GEB)

Department of Genetic Engineering and Biotechnology, University of Rajshahi

- **Field(s) of Study:** Advanced Molecular Biology, Advanced Plant Biotechnology, Genomics & Proteomics, Food Biotechnology, Nanobiotechnology, Advanced Biostatistics.
- **Thesis:** Sustainable Production and Molecular Characterization of Recombinant Invertase from *Saccharomyces cerevisiae* for Food and Pharmaceutical Applications **(Ongoing)**
- **Academic Progress:** First Semester (3.67 out of 4.00) and Second Semester (Result Awaited)

21/01/2019 – 07/10/2024

Bachelor of Science in Genetic Engineering and Biotechnology (B.Sc. in GEB)

Department of Genetic Engineering and Biotechnology, University of Rajshahi

- **Field(s) of Study:** Molecular Biology, Plant Science, Plant Pathology and Protection, Plant Biotechnology, Microbiology, Biostatistics, Plant Developmental Biology, Plant Breeding, Physiology and Biochemistry of Plants.
- **Project:** Safeguarding Betel Leaves: A Systematic Study on Salmonella Detection and Antimicrobial Mitigation Strategies
- **Result:** 3.57 out of 4.00

Core Qualifications

- **Wet-Lab:** Genomic and Plasmid DNA Extraction, Target Gene Selection, Codon Optimization (host-specific), Primer Design, Gene Amplification, Gel Electrophoresis, DNA Quantification, Restriction Digestion, Ligation, Preparation of Competent Cells, Transformation etc.
- **Wet-Lab Tools Skills:** PCR Thermal Cycler, Centrifuge, Homogenizer, Sonicator, UV-Vis Spectrophotometer, NanoDrop etc.
- **Microbiology Lab Skills:** Microbial Culture, Isolation & Characterization of Bacteria and Fungi, Bacteria, Colony Counting, Store Bacterial Strains, Gram Staining, Biochemical Testing, Antibiotic Susceptibility Testing etc.
- **Plant-Microbe Interaction Analysis:** Isolation of Pathogenic and Beneficial Microbes, Pathogen Identification, Molecular identification of the Microbes.
- **Dry-Lab:** Structure Based Drug Design, Virtual Screening, Molecular Docking, Structure Preparation etc.
- **Computational Chemistry and Bioinformatics Tools Skills:** ChemDraw, Gaussian, Pymol, AutodockVina, Chimera, Discovery Studio, YASARA etc.
- **Computer Skills:** Microsoft Word, Microsoft Power point, Microsoft Excel, Reference Management Software's (EndNote, Mendeley) etc.
- **Programming Language:** R Programming (Basic Level).

PERSONAL SKILLS

Mother tongue(s): Bengali

Other language(s)

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	C1	B2	B2	B2

Levels: A1/A2: Basic user - B1/B2: Independent user - C1/C2 Proficient user
Common European Framework of Reference for Languages

ADDITIONAL INFORMATION

International and National Conferences

- Islam, M.; **Hassan, MS**.; Hassan, R.; Mondol, M.; Hassan MA.; Johra, F.; Diba, NA.; Nasirujjaman, K.* "Isolation and Characterization of Rhizospheric Bacteria from Heavy Metal Contaminated Soil of Savar, Dhaka and Evaluation of Their Potential Plant Growth Promoting Attributes." International Biotechnology Conference 2025, hosted by BRAC University, Dhaka (https://ibc.bracu.ac.bd/).
- Islam, MS.; **Hassan, MS**.; Roqunuzzaman, M.; Shehab, MN.; Sujon, KM.; Reza, MA.* "Molecular Characterization of *PNPLA3* Gene Variants: PCR-Based Detection and Analysis of Novel Mutations in NAFLD Patients from Rajshahi Demographic".RURS 1st International Student Conference and Research Fair (ISCRF-2024), organized by Rajshahi University Research Society (https://rursbd.org/conference/).

- **Hassan, MS.**; Junaid M.* "A Conceptual Illustration of the Base Editing Interface." Single Figure Presentation (SFP-Equivalent to 3MT Presentation) organized by Virtual Winter School on Computational Chemistry 2023 (https://winterschool.cc/).

Submitted Articles in Journal

- Hossain, A.; Rahman, ME.; Jakia, N.; Rahman, MS.; **Hassan, MS.;** Sraboni, FS.; Emran, TB.; Nasirujjaman, K.* "Deep Learning-assisted Identification of Natural Plant-based PDE5 Inhibitors Targeting Erectile Dysfunction: Insights into Molecular Mechanics and Therapeutic Potential". (Computational Biology and Chemistry).

Completed Work: Manuscript Ready for Submission

- **Hassan, MS**.; Junaid, M.*; Siddika, A. "Leucas zeylanica: An Updated Review of Ethnomedicinal Uses, Phytochemistry, and Versatile Biological Activities".

Research Training

- 2024| "Introduction to R Programming for Data Analysis"(17th November- 18th December) at Institute of Environmental Science, University of Rajshahi.

  Learning: Effective data analysis in plant health, Analyze and visualize complex agricultural datasets, Statistical modelling to study plant health, Analyzing pest dynamics, Predictive Insights etc.

- 2024| "Agricultural Science-Soil Science and Plant Nutrition" (Self-paced) at Alison (CPD Certified) Empower Yourself.

  Learning: Planting pattern analysis, Pest and weed control, Seed germination factors, Propagation methods, Growth environment needs, Horticulture significance, Nursery establishment factors etc.

- 2023 | "PubChem Training Course" (24th -31st March) at National Library of Medicine (Network of the National Library of Medicine).

  Learning: Bioactive compound exploration for pest and disease management, PubChem resource utilization, Pesticide data interpretation, Chemical structure analysis, Research workflow integration etc.

- 2020-2021 | 7 days long Hands-on Training on ''Computer-Aided Drug Design: Desktop to Publication" (16th December-2nd January) at Advanced Bioinformatics, Computational Biology and Data Science Laboratory, Bangladesh (ABCD Laboratory, Bangladesh).

  Learning: Drug design principles and its applications in plant health, Optimize compounds for plant health, Eco-friendly agrochemical design, Molecular interaction analysis, Research publication integration etc.

Memberships

- **Undergraduate Member** | 07 September 2022 to Present |British Pharmacological Society.
- **Global Outreach- Student Member** | 22 September 2021 to Present |American Society for Microbiology.

Awards and Achievements

- **Champion** [Badminton-Double (Male)] and [Badminton- Single (Male)] in the Indoor Games 2024 Competition Organized by the Genetic Engineering and Biotechnology Society, University of Rajshahi.
- **Champion** [Badminton-Double (Male)] and **Runner-Up** [Badminton- Single (Male)] in the Indoor Games 2022 Competition Organized by the Genetic Engineering and Biotechnology Society, University of Rajshahi.

Social and Academics

- **Member** | December 2024 to Present | Rajshahi University Genetic Engineering and Biotechnology Alumni Association (RUGEBAA)

Volunteer

- **Executive Member** | February 2021 to April 2023 | Network of Young Biotechnologists of Bangladesh (NYBB)